Managing High Performance Scientific Computing Infrastructure

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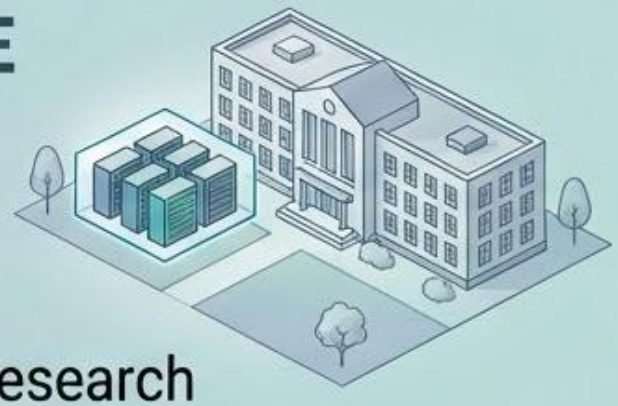
SCOPE OF THIS TALK

✕ OUT OF SCOPE

- Hyperscale cloud providers & National / flagship exascale systems
- Very different scales (100k+ nodes, 100+ MW)
- Dedicated, large SRE/DevOps teams, custom tooling under commercial or national-mission constraints

✓ IN SCOPE



- Institutional / research HPC facilities
- Typical scale: 10–100 racks, kW–single-digit MW, small teams

Focus: How to run these as **reliable, sustainable research platforms**.

From Ad-hoc Cluster to Research Platform

Ad-hoc Cluster

- Extremely Fast Growth for Computation
- Haphazard purchases
- No long term planning
- Bought opportunistically
- “hero admin” culture
- limited documentation

The Fixer Upper

Research Platform

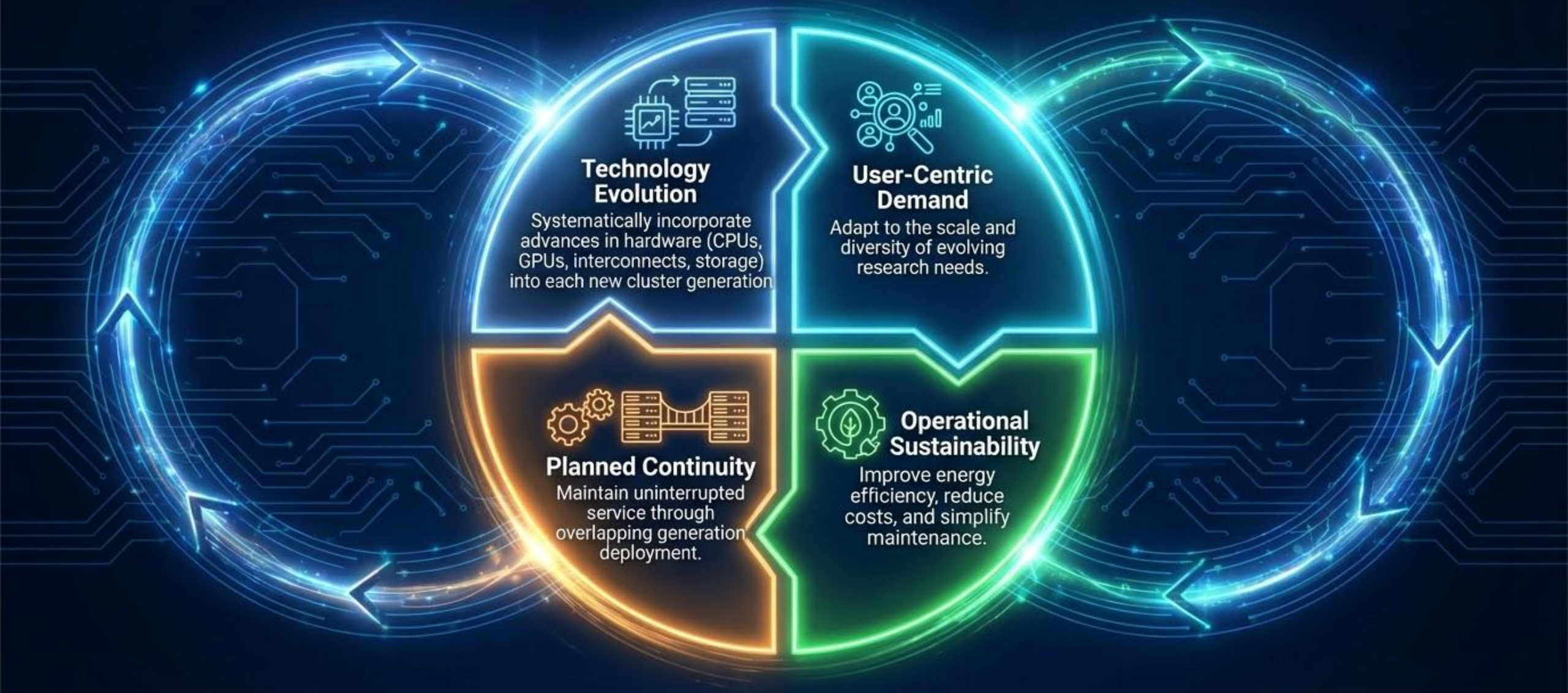
- Clear services
- Lifecycle plans for compute & storage
- Sustainability and cost built-in
- Telemetry-driven decisions

The Facility Planning

- **Long-Term Planning Cycles:** European university facilities often planned 5-10 years in advance.
- **Disconnected from Future Demands:** Planning fails to account for the rapid **explosion** in technology and **energy requirements**.
- **The Prediction Challenge:** Predicting exact needs for 7-10 years ahead is extremely difficult.
- **Outdated Planning Metrics:** Focus is often on rack count and floor space, disregarding critical metrics like power density (kW/rack) and future cooling needs.

Generational HPC Lifecycle Management

Key drivers of this lifecycle strategy include:



THE LIFECYCLE CONCEPT

Treat compute as a lifecycle, not a purchase

INTRODUCTION → GROWTH → PLATEAU → RETIREMENT

Typical cycle: 4–6 years, but shorter for GPUs/AI nodes

STRATEGIC BALANCE & INNOVATION

- Balance innovation vs stability
- Isolate "experimental" hardware from core production
- Use partitions/QoS to pilot new technologies

METRICS & END-OF-LIFE

KPIs for compute lifecycle

- Utilization (core-hours/GPU-hours vs capacity)
- Queue delays & job success rates
- Cost per core-hour / GPU-hour

Retirement options

- Decommission and recycle
- Reassign to low-priority / teaching / test clusters

Generation Planning Phases

Strategic Planning & Infrastructure Assessment

- Define lifecycle roadmap (Gen N, N+1, N+2)
- Audit current capacity and limitations
- Identify legacy systems to retire
- Needs assessment and analysis

Facility & Hosting Strategy

- Assess cooling, power, rack density
- Engage construction/IT on consolidation plans
- Evaluate colocation opportunities
- Model TCO scenarios

Architecture & Technology Planning

- Identify target technologies (CPUs, GPUs, fabric)
- Draft technical requirements for Gen N+1
- Begin vendor exploratory conversations

Financial Modeling & Budget Integration

- Define multi-year CapEx/Opex projections
- Align lifecycle with budget cycles
- Develop 3-tier funding scenarios (Core / Optimized / Stretch)

Stakeholder Alignment & Review

- Present lifecycle plan to institutional leadership
- Collect feedback from researchers
- Update risk log and approval dependencies

Governance & SOP Development

- Define version-controlled workflows
- Formalize documentation/ audit strategy
- Integrate automation, audit checkpoints, and training

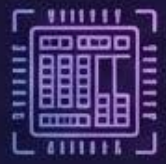
Gen N+1 Pre-Procurement Enablement

- Draft RFP/RFI criteria
- Finalize architecture for Gen N+1
- Conduct readiness reviews

Balancing AI & Classical Simulation: Storage and Compute

AI / ML: Storage for Many Small Files

- Millions of small files, checkpoints, preprocessed shards, logs, model versions
- Opposite storage tuning requirements
- Metadata-heavy, many parallel readers, lots of random I/O → benefits from:
  - High IOPS, fast metadata servers
  - Smaller stripe sizes or even "no wide striping"
  - SSD / NVMe tiers or object storage for training data



Classical Simulations: Storage for Few Large Files

- Comparatively few very large files (trajectories, grids, dumps, restart files)
- Mostly sequential streaming I/O → benefits from:
  - Large stripes across many disks
  - Wider RAID/stripe sets to maximize throughput
  - Careful tuning of RAID level, stripe size, and file layout

Practical Design Consequences:

- Separate "AI scratch / dataset" and "simulation scratch" where possible
- Different mount options and defaults (stripe count, stripe size) per filesystem / project type
- Encourage data lifecycle: hot AI datasets on fast tiers, cold training data and old results on cheaper capacity tiers

Lifecycle Management: Storage & Data

Compute is easy, data is hard

HOT STORAGE
(Scratch & Fast Parallel FS)

Short lifetime, high throughput. Default retention for scratch (days/weeks).

→ Data migration based on age and tags

WARM STORAGE
(Project / Working Storage)

Months–years. Project quotas + review cycles.

→ Data migration based on age and tags

COLD / ARCHIVE
(Object Storage, Tape, External Services)

Years+. Archive policies tied to grants/publications.

POLICIES & TOOLING

- Automated data expiry / migration
- Self-service reports for user visibility

Strategic question: What data is the HPC facility responsible for, and where does that responsibility end?

High Speed Interconnect

Traditional / "Enterprise" Networks

- Patchwork of hybrid meshes and leaf-spine
- Optimized for north-south traffic (client ↔ server)
- Oversubscription is normal; microbursts less critical
- Good enough for storage + login + small jobs

HPC Interconnect Requirements

- East-west, all-to-all, low-latency communication
- Predictable bisection bandwidth for large MPI jobs
- Tight coupling between scheduler placement and network topology
- Congestion and jitter directly impact time-to-solution

moving from "whatever the DC has" to purpose-built HPC network fabrics is key for scaling modern simulation + AI workloads.

Balancing AI & Classical Simulation (Continued)

Compute & Energy: Getting the Mix Right

- Power-aware scheduling as a first-class feature
- Apply power caps to GPU nodes
- Use energy-aware scheduling policies so jobs land where power and cooling are available
- Prefer "pack and power down" over "spread and idle" to reduce baseline consumption
- Retire or tame bad actors

Continuously measure perf-per-watt at node or partition level

- Downclock or reclassify the worst PUE / perf-per-watt hardware to low-priority queues
- Plan decommissioning based on efficiency, not just age
- Purchase for real workloads, not benchmarks
- Use telemetry (utilization, queue times, job profiles) to decide:
  - How many GPUs vs CPU cores you actually need
  - How much memory and storage bandwidth per node
  - Ignore "hero" benchmark numbers that don't match your users' codes
  - Optimize for performance per watt and performance per euro, not raw FLOPs

Telemetry & Observability

You can't manage what you can't see. At the same time, don't collect everything you can. Be strategic, otherwise the signal will be lost in the noise

Collect metrics from:

- **Hardware:** CPU/GPU utilization, memory, I/O, network, power
- **Scheduler:** job states, wait times, failures
- **Storage:** throughput, IOPS, capacity, metadata load
- **Facilities:** power feeds, PDUs, cooling, temperature

Build dashboards for different audiences

- **Admins:** detailed, low-level metrics for debugging
- **Management:** high-level utilization, reliability, cost trends
- **Users:** simple views of queue status, storage usage, incidents

Combine metrics with logs & traces

- Faster incident triage and post-mortems
- Evidence-based roadmap decisions

Telemetry for Capacity Planning & Incidents

From Graphs to Decisions: Capacity Planning

- Analyze utilization by project, partition, or workload type
- Identify “pain points”: overloaded queues, storage hotspots
- Use historical data for “what if” scenarios:
  - What if we double GPUs?
  - What if we retire this partition?

Incident Response

- Baseline normal behavior → detect anomalies
- Correlate events (e.g., power spikes, firmware updates, user campaigns)
- Shorter MTTR through targeted diagnostics

Telemetry → Narrative

- Turn metrics into stories for leadership:
  - “We are at X% capacity; here’s what happens if we don’t expand.”

Power, Cooling & Sustainability

Energy is now a first-class requirement

Driven by...

- Electricity prices
- Institutional climate targets
- Public perception

Design & Operate Efficiently

Design and operate with efficiency in mind:

- Choose efficient nodes (perf/Watt, not just peak FLOPs)
- Consider higher density / liquid cooling for GPUs
- Monitor PUE-like indicators where possible

Power-aware operations

- Schedule maintenance during low-demand periods
- Adapt to known power constraints or curtailment events

Communicate the Impact

- kWh and CO_2 equivalents per year
- Efficiency improvements across generations

Thank You

Questions?

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